

Team Meeting Minutes: October 6, 2026

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Meeting details

Field	Value
Date	October 6, 2026, about 4:04 PM
Length	About 34 minutes
Attendees	Student team: Tyler, Joe and Chris
Sponsor	Dr. Qualls not present
Purpose	Team work session: thresholds and precompute, real-data demo, VIIRS, schedule, and budget

User-defined thresholds and precomputing

The meeting picked up mid-discussion about how much the dashboard should let users change.

- One idea from an earlier meeting came up again: let the end user choose the snow-removal model, with a disclaimer saying which version produced the results and that they are not 100% accurate.
- The team recalled that Dr. Qualls wants the threshold to be user-defined, though not necessarily as many options as the team had been considering. Watershed selection is part of the project regardless.
- The code should be modular, both so thresholds can be changed through user input and because the long-term goal is worldwide coverage. It can be scaled back later if needed.
- Precomputing makes threshold changes fast, at least for the default threshold. The open question is storage: offering thresholds in 10% steps would mean about ten copies of the data. That is manageable for Idaho but grows quickly if the tool expands.
- A team member wants to run tests on how fast the FDL step, the intensive part, can be made.
- Someone recalled that the starter code has an error that quadruples the data size (see the corrections below).

Demo: real MOD10A1F data

Joe had an AI coding assistant build a real-data version of the demo. It downloads NASA's cloud-gap-filled MODIS files (MOD10A1F) and runs the full pipeline. He pushed it to the repo during the meeting (`realdata/`, commit "add real data working demo").

- **FDL on just the watershed took about 2 seconds.** Masking to the watershed before running FDL, instead of running it on the whole state, is one way to support a custom threshold: the user picks the watershed first, then FDL runs only there.
- The FDL code is the starter code with its known issues fixed. It was checked against the original starter code on dummy data and gave the same result.
- The raw data is gitignored because it is many gigabytes. The `realdata/` README explains how to download it.
- Timings reported: reading one tile-year of files about 1.7 seconds; FDL for a full tile about 5 seconds; FDL for 25 years about 6 minutes. A one-time 25-year Idaho precompute is about 40 minutes, mostly downloading, then one new year each spring.
- **Can we query an API instead of downloading?** No. There is no API that returns just the pixels needed. The daily files have to be fetched, but they do not have to be kept: download once for the precompute, then once a year.

VIIRS data

- Tyler posted screenshots in the chat. What he uploaded also pulls VIIRS data.
- MODIS and VIIRS look close to one to one. He found a small correction for shifting the pixels, but they show generally the same thing.
- He also tried combining the two into a composite: where one sensor has cloud over a pixel and the other has a clear view, the clear one fills it in, giving a fuller picture.
- The team liked the idea but agreed it is outside the scope and should not create extra work now. Dr. Qualls asked for a MODIS or VIIRS option, not a combination. It could be offered later as an extra.

Schedule and course deadlines

- The schedule looks good and lines up with the design validation plan. Nobody went through every line.
- Someone noticed the schedule may have had a sponsor deliverable due today or yesterday. This needs checking.
- Idea: import the schedule into GitHub so it becomes a list of tasks. GitHub accepts it, though the spreadsheet's layout makes it awkward.

- **Snapshot Day 1 is October 20:** storyboard presentations, possibly over Zoom. More details on October 13, the first day back in class.
- **The design expo is in the spring** (late April) and needs a poster. The team expects to travel down for it. Technical presentations start at 1:20 PM on the first day, and the second day runs from about 9 AM to 6 PM, so it is one night away.
- Assignments 14 and 15 cover the midterm material.
- Completed assignments go in the project planning folder on OneDrive.

Budget

The total is about \$1,500. The expo trip, at about \$600 to \$700, is the most expensive part, and the department may already have arrangements for it. The budget can be updated as the project goes.

- **Expo travel.** The template already listed \$550 to drive down in April, which needs to go up to cover lodging and meals.
 - Lodging: set to \$250 per room per night. The sheet's formula already multiplies by two rooms. One night is needed.
 - Alternatives looked at: a Moscow townhouse at \$389 for the night, and other rentals at about \$242 to \$389. Hotels came out about the same.
 - Meals: \$75 per team meal for four people.
- **Storage.** The sheet first priced a 2 TB portable SSD at \$150. A 2 TB portable SSD costs about \$300 or more, so it was updated to \$300. The cell's note estimates 4 MODIS tiles × 26 years × 180 days × about 10 MB per file, roughly 200 GB of raw data, plus the FDL layers (see the corrections below).
- A leftover note in the total cell, from the template, named someone outside the team. The team planned to remove the template notes before submitting.
- **Licenses:** none needed. The databases and front-end tools the team expects to use are open source.

Corrections

1. Threshold copies may not be needed. At the October 1 meeting, Dr. Qualls said the default is 10, with an advanced option to change it. He also said a PCA built at 10 can cloud gap fill at any user threshold up to about 50 (15, 30, 38.5 and so on) without recomputing. So user-chosen gap-fill thresholds do not need extra stored copies. Only an advanced user changing the FDL threshold itself would, and those could be offered at a few fixed values.

2. The starter-code error affects memory, not stored size. The original script decodes the uint8 MODIS values to 32-bit floats, four times the size. That makes one tile-year about 8 GB in memory

instead of about 2 GB, and it is also what turned fill and missing pixels into FDL = 0. The rewritten code reads raw uint8, so both problems are fixed. The files written to disk were never affected.

3. Tiles. The September 22 notes said Idaho is covered by four tiles, and this meeting said two. Both are partly right. The state outline fits in two tiles (h09v04 and h10v04). But checked against the USGS boundaries on October 6, the 92 HUC8 watersheds that touch Idaho need **three**: h09v04 (74 watersheds), h10v04 (33) and h10v03 (2, near Canada). 17 of them span two tiles.

4. The storage estimate is too high for what needs to be kept. Measured from NASA's catalog on October 6, a MOD10A1F file is about 2 to 4 MB, not 10 MB, and the 3 tiles above are enough. More importantly, the raw files do not need to be kept after FDL is computed:

What is stored	3 tiles, full record
Precomputed FDL and LDS rasters only (MODIS and both VIIRS)	About 4 GB or less
Raw gap-filled files (MOD10A1F, VNP10A1F, VJ110A1F)	About 184 GB
Raw files including the cloudy products	About 610 GB

If only the FDL rasters are kept, which matches the precompute plan, the SSD line may not be needed at all.

5. Timing context. The demo timings were measured on a 14-core Mac. The 1.7-second read was probably helped by the files already being in the operating system's cache, so a first run will be slower.

Action items

Owner	Task	Due
Team	Check which sponsor deliverable the schedule lists as due October 5 or 6, and whether it is done	Now
Unassigned	Upload the budget to the project planning folder on OneDrive (offered at the end of the meeting)	Now
Team	Move the other completed assignments into the project planning folder	Now
Team	Revisit the storage line in the budget using the measured sizes, and remove the template notes	Before submitting
Team	Find out whether Snapshot Day 1 on October 20 is in person or on Zoom	October 13

Team	Confirm the expo dates and whether the department arranges travel or lodging	Not set
Team	Review assignments 14 and 15 for the midterm	Before the midterm
Team	Benchmark the FDL step: per watershed vs. per tile, and the cost of each extra threshold	Not set
Team	Decide which thresholds to offer and precompute, and confirm with Dr. Qualls	Open
Tyler	Write up the VIIRS pixel-shift correction. Note that VIIRS is on a 375 m grid and MODIS on 500 m, so the two cannot be stacked without resampling.	Not set
Joe	Push the real-data demo and its README	Done